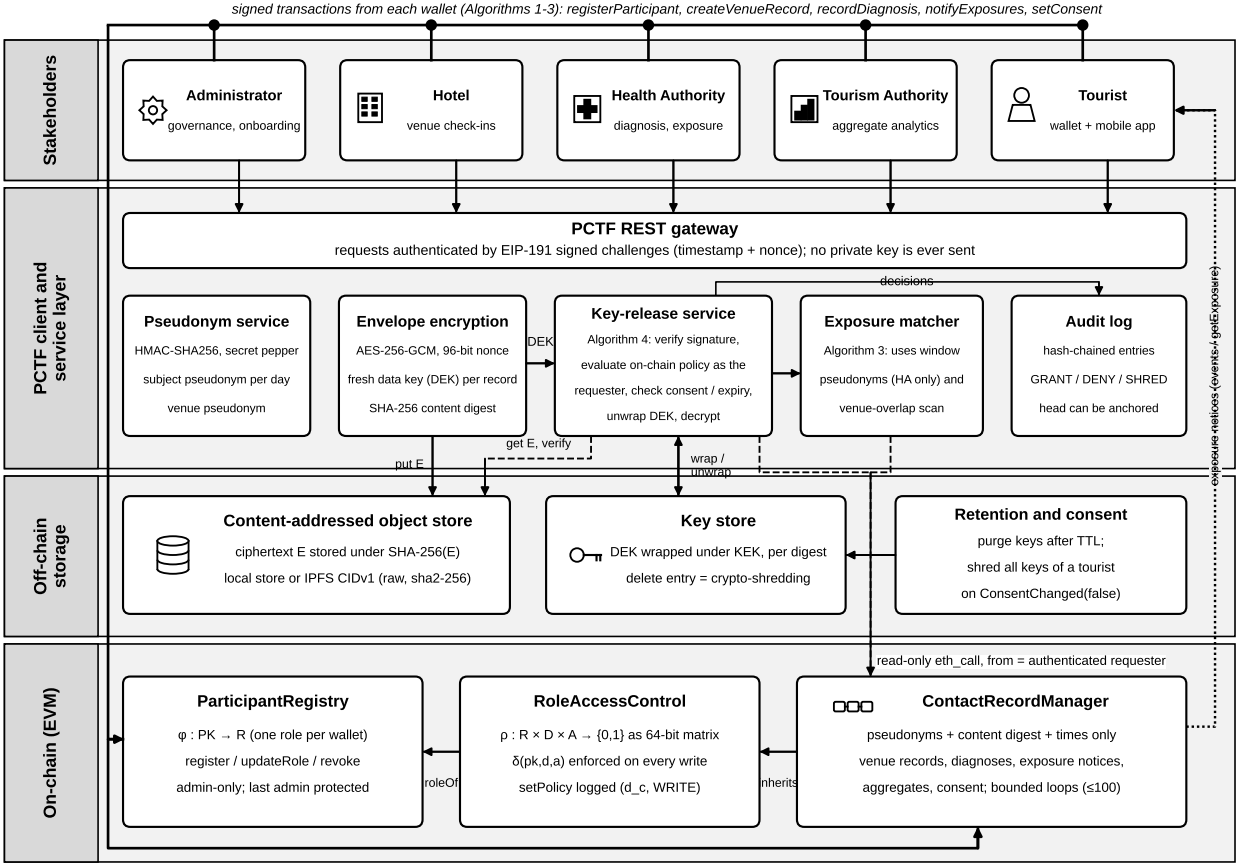

signed transactions from each wallet (Algorithms 1-3): registerParticipant, createVenueRecord, recordDiagnosis, notifyExposures, setConsent
Stakeholders
Administrator
governance, onboarding
Hotel
venue check-ins
Health Authority
diagnosis, exposure
Tourism Authority
aggregate analytics
Tourist
wallet + mobile app
PCTF client and service layer
PCTF REST gateway
requests authenticated by EIP-191 signed challenges (timestamp + nonce); no private key is ever sent
decisions
Pseudonym service
HMAC-SHA256, secret pepper
subject pseudonym per day
venue pseudonym
Envelope encryption
AES-256-GCM, 96-bit nonce
fresh data key (DEK) per record
SHA-256 content digest
DEK
Key-release service
Algorithm 4: verify signature,
evaluate on-chain policy as the
requester, check consent / expiry,
unwrap DEK, decrypt
Exposure matcher
Algorithm 3: uses window
pseudonyms (HA only) and
venue-overlap scan
Audit log
hash-chained entries
GRANT / DENY / SHRED
head can be anchored
exposure notices (events+ getExposure)
get E, verify
put E
wrap / unwrap
Off-chain storage
Content-addressed object store
ciphertext E stored under SHA-256(E)
local store or IPFS CIDv1 (raw, sha2-256)
Key store
DEK wrapped under KEK, per digest
delete entry = crypto-shredding
Retention and consent
purge keys after TTL;
shred all keys of a tourist
on ConsentChanged(false)
read-only eth_call, from = authenticated requester
On-chain (EVM)
ParticipantRegistry
φ : PK → R (one role per wallet)
register / updateRole / revoke
admin-only; last admin protected
roleOf
RoleAccessControl
ρ : R × D × A → {0,1} as 64-bit matrix
δ(pk,d,a) enforced on every write
setPolicy logged (d_c, WRITE)
inherits
ContactRecordManager
pseudonyms + content digest + times only
venue records, diagnoses, exposure notices,
aggregates, consent; bounded loops (≤100)